



The mirrored arterial dilemma

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ABSTRACT

An 89-year-old man presented with acute and painless loss of vision in his left eye. He denied constitutional symptoms. Visual acuity was light perception with pallid optic disc edema. Neurological work-up and serological inflammatory markers were normal. Due to the clinical suspicion for giant cell arteritis (GCA), he was started on corticosteroid therapy and bilateral temporal artery biopsies (TABs) were performed. Histopathology of the left temporal artery ipsilateral to the visual loss was negative for arteritis, while the contralateral temporal artery was positive for arteritis. The diagnostic evaluation of GCA and value of unilateral versus bilateral TABs is discussed.

1. Case report

An 89-year-old man with a history of cataract surgery both eyes (OU), arterial hypertension, coronary artery disease, aortic valve replacement on anticoagulation, and excision of a squamous cell carcinoma of the right lower eyelid presented with 4 days of rapidly progressive vision loss in the left eye (OS). He reported intermittent mild retroorbital pain OU, but denied any trauma, recent illnesses, fevers, weight loss, jaw pain with chewing, or joint pain. Initial neurological and serological workup was negative at an outside facility. In our Emergency Department, visual acuity was 20/30 right eye (OD) and light perception OS, with a relative afferent pupillary defect OS. There scattered macular drusen with unremarkable retinal vessels and periphery OU. The right optic disc appeared normal with a cup-to-disc ratio of 0.3. There was left optic disc edema with inferior pallor (Fig. 1). Blood pressure was 148/66 mmHg with a normal erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP).

At this point given the patient's findings, do you have any concerns?

Is there any physical examination or laboratory findings that could help point to the diagnosis?

2. Comments by M. Tariq Bhatti, MD

Before I comment on my working diagnosis and what I am most concerned about, there are several important items I need to know:

- What medications is the patient taking? For example, amiodarone, phosphodiesterase type 5 inhibitors, and immune checkpoint inhibitors have all been associated with optic disc edema.
- What was the result of the complete blood count (CBC)? An abnormal white blood cell count could suggest an infectious or neoplastic process. If the hematocrit is less than 35 %, the ESR needs to be corrected using the following formula: $ESR \times 15 / (55 - \text{hematocrit})$.¹⁰ An elevated platelet count ($> 400,000/\mu\text{L}$) could represent an inflammatory process such as giant cell arteritis (GCA).²²
- Is there tenderness, nodularity, beading or pulselessness on palpation of the superficial temporal arteries to suggest GCA?

From what we are told, I think the most likely diagnosis at this time—given the lack of systemic symptoms with normal ESR and CRP—is non-arteritic anterior ischemic optic neuropathy (NAION), but there are at least 5 caveats to consider in this patient:

1. **light perception vision.** Although such profound vision loss can occur in NAION, it should give pause to consider the possibility of arteritic anterior ischemic optic neuropathy (AAION) due to GCA. In the Ischemic Optic Neuropathy Decompression Trial, only 21 % of the patients enrolled with NAION had a vision of counting fingers to no light perception.³⁰ Furthermore, in a study of 237 eyes with NAION, seen within 2 weeks of onset of visual loss, only 14 % had a visual acuity of counting fingers or worse.¹⁶ In comparison, the visual loss in GCA is often profound, with approximately 88 % of patients having a vision of 20/200 or worse.²³

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2. **pallid optic disc edema.** It appears to me based on the photograph that the patient has pallid optic disc edema, which has been classically associated with GCA, but as reported recently that is not always the case.²⁰
3. **ESR and CRP.** There is no one single blood test that is 100 % sensitive for GCA. The sensitivity of CRP and ESR is approximately 86.9 % and 84 %, respectively; and 99 % for combined ESR and CRP.^{12,31}
4. **occult GCA.** A clinical variant of GCA known as occult GCA (defined as GCA without any systemic symptoms) occurs in approximately 20 % of cases.¹⁴
5. **pre-temporal artery biopsy (TAB) probability of GCA.** When deciding on whether to do or not to do a TAB it is helpful to determine the pre-test probability of a positive result. For this patient, using a simple pre-TAB algorithm there would be a very low clinical suspicion for GCA (Fig. 2).⁹ Using the Ing GCA predication model,¹¹ the approximate risk of GCA for this patient is only 4.74 %.

In summary, I would say there are more reasons for this patient to not have GCA than to have GCA, but out of an abundance of caution and keeping in mind the caveats listed above, I would start corticosteroids and obtain a TAB just to be sure I am not wrong.

3. Case report (continued)

Given the patient's age, presenting complaint, and constellation of findings, there was concern for GCA. The patient was started on 60 mg of oral prednisone daily, and bilateral TABs were performed with 2 cm long segments obtained on each side. Histopathology of the left temporal artery demonstrated intimal hyperplasia and patchy disruption of the elastin layer with no active arteritis. Histopathology of the right temporal artery demonstrated patches of moderate to severe transmural chronic inflammation, including giant cells, with severe damage to the elastin layer of the vessel in the affected area consistent with the diagnosis of GCA.

What is the role of TAB in patients with concern for GCA? What are key histopathologic findings?

What are considerations for TAB, including the timing of biopsy relative to initiation of steroids, the goal length of the biopsy specimen, and when to proceed with unilateral versus bilateral biopsy?

4. Comments by Dr. Bhatti continued

Being a neuro-ophthalmologist, I rely on the pathologist to provide me the appropriate interpretation of the TAB specimen. Recently, a consensus paper on the processing, interpretation, and reporting of TABs

was published.²⁹ Twenty-one pathologists agreed on the following:

- Submit TAB specimens in total, uncut, for tissue processing, then cut them into 1–2 mm interval sections at the time of embedding.
- Review at least 5 slides per specimen, with up to 8 tissue profiles per slide.
- Active arteritis is a histologic diagnosis of any degree of inflammation involving the media of any artery represented on the slide and should not be made on the basis of immunohistochemistry alone.
- Active arteritis in a temporal artery is not synonymous with GCA, as many forms of arteritis can involving the temporal artery.
- The term “healed arterial injury” is favored rather than “healed arteritis.”
- Avoid using the terms “temporal arteritis” and “atypical arteritis.”
- Histological stains to be used include hematoxylin and eosin, elastic stain, or Movat stain and Congo red stain if amyloidosis is suspected.
- Consider immunohistochemical stains for histiocytes (CD68 or CD163) and T-lymphocytes (CD3) to delineate subtle involvement of the arterial wall by inflammation.
- Histological features include: lymphohistiocytic inflammation with disruption of the media, lymphocytic inflammation in the adventitia and periadventitial small vessels, disruption of the internal elastic lamina, intimal thickening, and Langhans-type giant cells, often along the internal elastic lamina (giant cells need not be present).

Book 5 (Neuro-ophthalmology) of the American Academy of Ophthalmology Basic and Clinical Science Course states that a TAB should be performed within 10–14 days from starting corticosteroid treatment.² I recommend obtaining at least a 2 cm segment of the temporal artery, but there are studies that suggest a longer segment is preferred, and other studies that have concluded a shorter segment is adequate.^{26,27} The answer to the question of whether to perform a unilateral or bilateral (simultaneous or sequential) TAB is not a simple one.¹ Many studies have shown that in approximately 5 % of cases a TAB can be negative on one side, but positive on the other side.⁴ I personally have rarely performed bilateral simultaneous TABs. In the vast majority of cases I have pursued a unilateral TAB, and if negative, proceeded with a TAB on the opposite side only if the patient had a high pretest probability for GCA. The American College of Rheumatology (ACR) still considers a TAB the gold standard for confirming GCA in the USA²⁴; but in other parts of the world such as in Britain and Europe a temporal and axillary artery ultrasound, high-resolution magnetic resonance imaging (MRI) or positron emission tomography (PET)/computed tomography (CT) are acceptable first-line diagnostic tests.^{8,25}

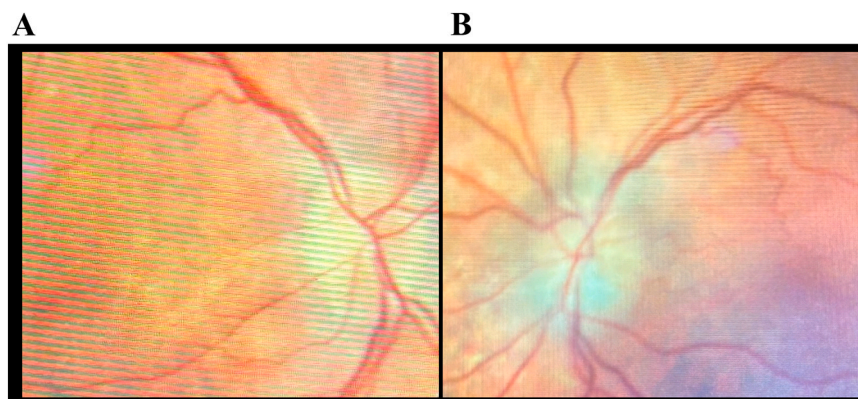


Fig. 1. A, 1B. Fundus image (OptoMed Aurora) of the right optic disc (left panel) and left optic disc (right panel) on the day of presentation to our institution. The right optic disc is well perfused with no edema. The left optic disc demonstrates inferior edema and pallor.

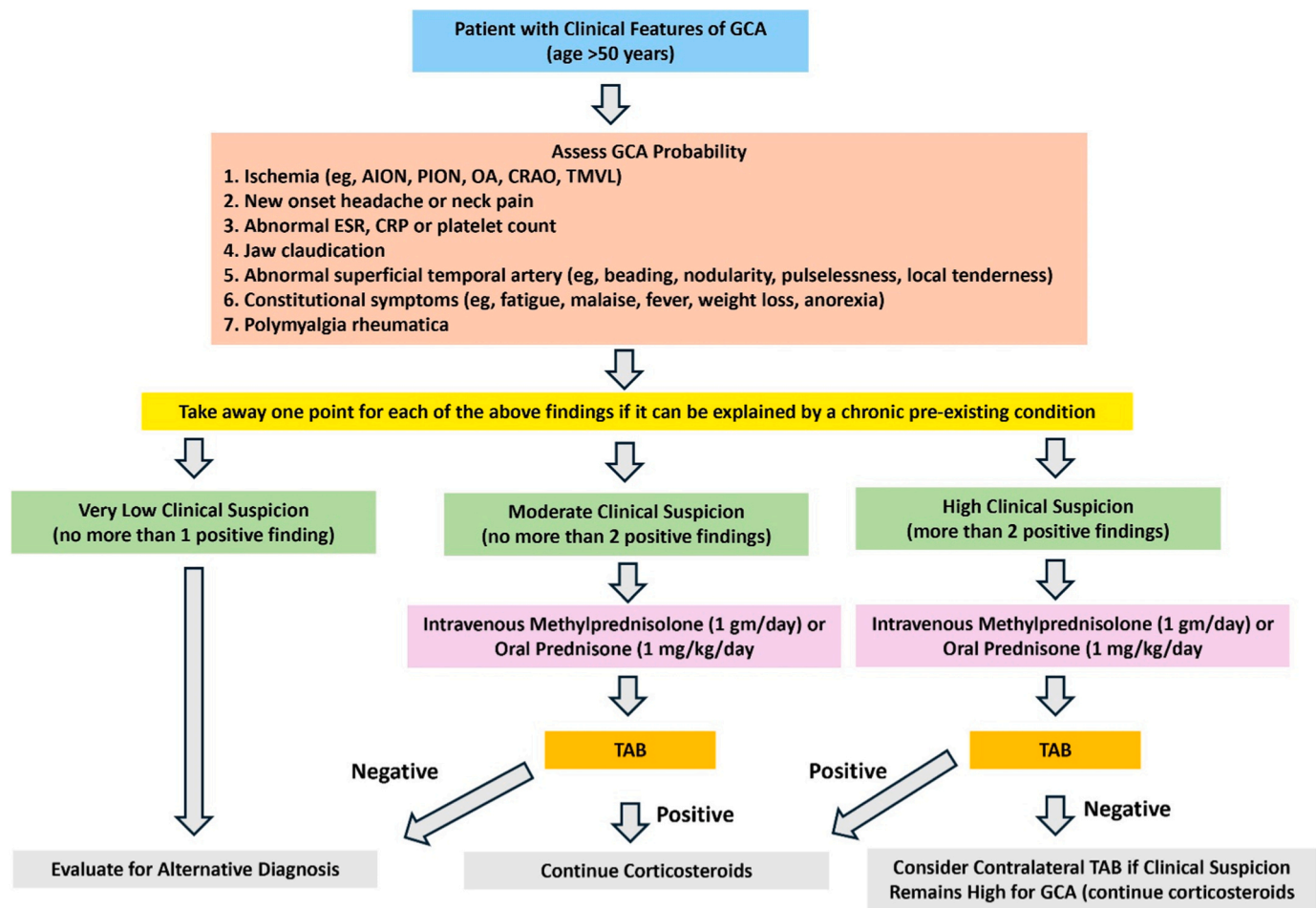


Fig. 2. Algorithm to determine pre-temporal artery biopsy risk for giant cell arteritis. Reproduced with permission from El-Dairi MA, Chang L, Proia AD, Cummings TJ, Stinnett SS, Bhatti MT. Diagnostic Algorithm for Patients With Suspected Giant Cell Arteritis. J Neuroophthalmol. 2015 Sep;35(3):246–53. Abbreviations: AION - anterior ischemic optic neuropathy; CRAO - central retinal artery occlusion; CRP - C-reactive protein; ESR - erythrocyte sedimentation rate; GCA - giant cell arteritis; OA - ophthalmic artery occlusion; PION - posterior ischemic optic neuropathy; TAB - temporal artery biopsy; TMVL - transient monocular vision loss.

5. Case report (continued)

Despite initiation of corticosteroids, the patient’s vision worsened to no light perception OS, and he reported subjective waxing and waning vision OD. He was admitted to the hospital and started on intravenous (IV) methylprednisolone 250 mg every 6 hours. MRI of the brain demonstrated chronic hypertensive changes with no acute changes. He received IV tocilizumab while hospitalized and was discharged on

100 mg of oral prednisone daily.

On follow-up 2 months after presentation, his vision remained stable OD and no light perception OS. There was diffuse optic disc pallor with resolved edema OS (Fig. 3). Surveillance CT angiography of the chest revealed a 12 mm splenic artery aneurysm and mild thickening of the aortic segments without clear arteritic changes. The corticosteroids were tapered, with the plan to receive monthly tocilizumab infusions over a year. Additionally, he was prescribed trimethoprim-sulfamethoxazole

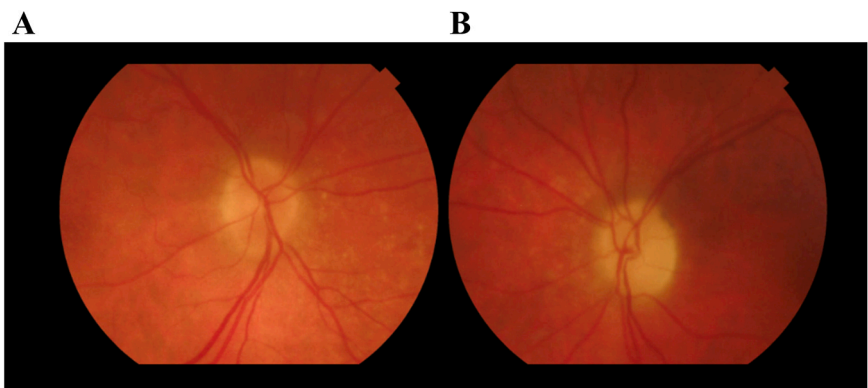


Fig. 3. A, B. Photographs (Optos) on follow up examination. The right optic disc appears unchanged and normal. There is left diffuse optic disc pallor without edema.

for *Pneumocystis jiroveci* pneumonia prophylaxis given his immunosuppressive state.

6. Discussion

GCA is a medium to large vessel inflammatory vasculitis that predominantly affects individuals over the age of 50 years. Cranial manifestations include headache, tongue necrosis, and jaw claudication; with one of the most feared complications being rapidly progressive irreversible vision loss largely due to AAION. In 1990, The ACR published risk stratification criteria for GCA that comprises a set of epidemiological, clinical, laboratory, and histopathologic findings.¹⁸ These include age at onset of symptoms at greater than 50 years of age, onset of a new headache, temporal artery tenderness or reduced pulsation, elevated ESR (>50 mm/h), and histological evidence of necrotizing arteritis in the temporal artery, with predominantly mononuclear cell infiltration or granulomatous inflammation (typically with multinucleated giant cells). Satisfying at least 3 of these 5 criteria is highly suggestive of GCA, with diagnostic sensitivity and specificity of 93.5 % and 91.2 %, respectively.¹⁸ Revised criteria in 2022 stratify the criteria via a points system and additionally include “halo sign” on temporal artery ultrasound, bilateral axillary involvement, and PET/CT activity of the aorta. A score of 6 points or greater is required to classify the disease as GCA.⁶

As such, initial diagnosis is typically clinical, with rapid initiation of high dose corticosteroids in suspected disease due to its vision-threatening nature. Since TAB is still considered the gold standard for the diagnosis of GCA in the USA, it should be performed urgently and preferably within 2 weeks of starting steroids and should not delay the start of therapy; however, case reports and case series have demonstrated that obtaining a biopsy after 2 weeks is still diagnostically useful.^{13,28,32}

Treatment of GCA often consists of a slow taper of systemic corticosteroids, typically prednisone, over 12–18 months. A more rapid taper can be considered in patients who are at high risk of severe complications from prolonged steroid use, such as patients with diabetes mellitus or who are immunocompromised. Although many clinicians opt for starting patients on high-dose IV corticosteroids, nonrandomized, largely retrospective studies have shown no statistically significant difference in final visual acuity between those started on oral corticosteroids versus those receiving pulse IV corticosteroids.^{15,17,19} In cases of relapse during taper therapy, adjunctive therapy with tocilizumab, an interleukin-6 inhibitor, is recommended. In patients who cannot tolerate tocilizumab, methotrexate can be considered.⁵ Steroid therapy is often guided by the biopsy results but may still be administered despite negative temporal artery biopsy if high clinical suspicion remains. Nevertheless, optimizing biopsy accuracy is important given the risks associated with prolonged corticosteroid use including osteoporosis, immunosuppression, psychological disturbances, or glycemic changes.⁵

Our case was unusual for several reasons. First, the inflammatory markers (ESR and CRP) were normal throughout the patient's clinical course, despite later having biopsy-proven GCA. Studies have shown that only a small percentage of corticosteroids-naïve patients with GCA (0.8–4 %) have a normal ESR and CRP at time of initial diagnosis.^{21,31} Second, the presenting symptom of vision loss was on the left side; however, active arteritic inflammation was only found in the right temporal artery. A retrospective study looking at TAB specimens submitted between 1968 and 1996 found that up to 94 % of bilateral or sequential TABs were concordant, and unilateral TAB had a high probability of yielding the correct diagnosis.³ Another retrospective study analyzing 91 patients who underwent bilateral simultaneous or sequential TABs found a concordance rate of 99 % among positive biopsy results⁷; however, TAB may be discordant with symptom localization as in our patient. The decision to perform unilateral, bilateral simultaneous, or bilateral sequential TAB is clinician and case dependent.

In conclusion, we present an uncommon case of seronegative GCA

with biopsy-positive disease present only on the contralateral side of symptoms, highlighting the usefulness of bilateral TABs in certain patients.

CRedit authorship contribution statement

Marissa K. Shoji: Writing – review & editing. **Skenda Jean-Charles:** Writing – original draft. **M. Tariq Bhatti:** Writing – review & editing, Writing – original draft. **Don O. Kikkawa:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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